

| Question |     |    |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |          |          |           |          |          |
|----------|-----|----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          | (c) |    |  | 1. (Potassium ions/ malate) reduce water potential in (guard) cell (1)<br>Accept osmotic pressure increases/ osmotic potential decreases/<br>solute potential decreases/ hypertonic to outside<br>2. water moves in by osmosis (down water potential gradient) (1)<br>3. (Turgor) pressure inside (guard) cell increases/ cells become turgid<br>(1) NOT cells expand<br>4. ends of guard cell have a thinner wall than centre/ ORA (1)<br>5. ends of guard cell expand and stomata opens (1) | 3               |          | 2        | 5         |          |          |
|          | (d) | i  |  | $2 \times \pi \times 2 \times 423 = 5312.9$ (1)<br>$= 5310$ ( to 3 sig figs) (1)<br>Allow 5320 if they use value of $\pi$ from calculator.<br>5310/ 5320 = 2 marks<br>5312.9/ 5313/ 5315.6/ 5316 = 1 mark<br>Evidence of $2 \pi r \times 423 = 1$ mark                                                                                                                                                                                                                                        |                 | 2        |          | 2         | 2        |          |
|          |     | ii |  | (water molecules) escape more readily from) species B because it<br>has larger (total) circumference.<br>Ecf if calculation incorrect in (i)                                                                                                                                                                                                                                                                                                                                                  |                 | 1        |          | 1         |          |          |
|          |     |    |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>5</b>        | <b>7</b> | <b>2</b> | <b>14</b> | <b>2</b> | <b>3</b> |